University of Central Punjab

Faculty of Information Technology

Object Oriented Programming

Fall 2025

Lab 04	
Topic	Classes in C++, access specifiers, objects, methods, constructors, destructor.
Objective	Making students familiarize with the concepts of constructor, overloaded constructor, getter and setter functions and classes.
CLO	Apply object-oriented programming principles to implement real-world problems.

Instructions:

- Indent your code.
- Comment your code.
- Use meaningful variable names.
- Plan your code carefully on a piece of paper before you implement it.
- Name of the program should be same as the task name. i.e. the first program should be Task_1.cpp
- **void main() is not allowed. Use int main()**

- **You are not allowed to use any built-in functions**
- **You are required to follow the naming conventions as follow:**
    - **Variables:** firstName; (no underscores allowed)
    - **Function:** getName(); (no underscores allowed)
    - **ClassName:** BankAccount (no underscores allowed)

Students are required to complete the following tasks in lab timings.

Task 1:

Create a class named as **ComplexNumber** having following private attributes:

- **realPart(double)**
- **imaginaryPart(double)**

Now write the following functions for the above-mentioned class:

a) The program should include setter function(s) to assign user defined values to the above-mentioned variables.

void setValue()

b) The program should include getter function(s) to get values for the above-mentioned variables.

void getValue()

c) Write a **display** function to print the attributes of the class in the format given below.

void display()

d) Write a program to create **two objects** of **Complex Number** with different data. And then display all the complex number objects in the following format.

e) Write a function **sum** which calculates the sum of two complex numbers.

f) Write a function **difference** which calculates the difference of two complex numbers.

Sample Outputs:

1st Complex Number :
Enter Real part of Complex Number: 6
Enter Imaginary part of Complex Number: 3

2nd Complex Number:
Enter Real part of Complex Number: 25
Enter Imaginary part of Complex Number: -30

1st Complex Number: 6 + 3i
2nd Complex Number: 25 - 30i

Sum of (6 + 3i) + (25 - 30i) = 31 + -27i
Difference of (6 + 3i) - (25 - 30i) = -19 + 33i

Task 2:

Design a class **Time** having following private attributes:

- **hour(int)**
- **minute(int)**
- **second(int)**

Now implement the following member functions for the above mentioned class:

a) The program should include **setter** function(s) to assign user defined values to the above-mentioned variables.
b) The program should include **getter** functions to get values for the above-mentioned variables.

c) Write a function **addTime(Time t)** to add two Time objects.

d) Write a function **display()** to display the time in the format **HH:MM:SS**

Sample Outputs:

Time 1: 12:45:30
Time 2: 2:30:45
Total Time: 15:16:15

Task 3:

Write a **Circle** class that has the following private member variables:

- **radius** (double)
- **pi** (double) initialized with the value 3.14159

The class should have the following public member functions:

a) Write the default constructor which that sets radius to 0.0.
b) Write the parameterized constructor which accepts the radius of the circle as an argument.
c) Define **setRadius** function for the radius variable.
d) Define **getRadius** function for the radius variable.
e) Define function **Area** that returns the area of the circle.
f) Define **getDiameter r**eturns the diameter of the circle.
g) Define a function **getCircumference** which returns the circumference of the circle.
h) Define a **display** function that displays the radius, area, diameter and circumference of the circle object.

Write a program to demonstrate the Circle class by asking the user for the circle's radius, creating a Circle object, and then displaying the circle's area, diameter, and circumference.

Task 4:

Create a class **Employee** with the following attributes:

- **name (char *)**
- **employeeID (int)**
- **salary (double)**

Implement the following member functions:

a) Write the default constructor which that sets above attribute to **null, 0** and **0.0**.
b) Write the parameterized constructor which accepts the employeeID as an argument.
c) The program should include **setter** function(s) to assign and update the values of the above-mentioned attributes.
d) The program should include **getter** functions to get values for the above-mentioned attributes.
e) Create **giveRaise** function that takes percentage as parameter, and then calculates the increased salary according to given percentage.
f) Write a **display()** function to display the employee details and the updated salary.

Sample Outputs:

Name:	**Alice**
Employee ID:	**101**
Salary:	**50000**
Giving a raise:	**10%**
New Salary:	**55000**

Task 5:

Now create a menu based program where user should select the function that needs to be tested from the above functions.

For example:

- Press 1 for Testing **‘ComplexNumber’** class.
- Press 2 for Testing **‘Time’** class.
- Press 3 for Testing **‘Circle,** class.
- Press 4 for Testing **‘Employee’** class.
- Press 0 for exit.